

-- 1. Create Tables

```
CREATE TABLE Customer (  
    Cust_id VARCHAR(10) PRIMARY KEY,  
    Fname VARCHAR(20) NOT NULL,  
    Lname VARCHAR(20) NOT NULL,  
    Area VARCHAR(10) NOT NULL,  
    Phone VARCHAR(15)  
);  
  
CREATE TABLE Movie (  
    Mv_no INT PRIMARY KEY,  
    Cust_id VARCHAR(10),  
    Title VARCHAR(50) NOT NULL,  
    Star VARCHAR(10) NOT NULL,  
    Price INT CHECK (Price BETWEEN 100 AND 250),  
    FOREIGN KEY (Cust_id) REFERENCES Customer(Cust_id)  
);
```

-- Insert Sample Data

```
INSERT INTO Customer (Cust_id, Fname, Lname, Area, Phone) VALUES  
( 'A01', 'Ivan', 'Ross', 'SA', '6125467'),  
( 'A02', 'Vandana', 'Ray', 'MU', '5560379'),  
( 'A03', 'Pramada', 'Jauguste', 'DA', '4560389'),  
( 'A04', 'Basu', 'Navindi', 'BA', '6125401'),  
( 'A05', 'Ravi', 'Shridhar', 'NA', NULL),  
( 'A06', 'Rukmini', 'Aiyer', 'GH', '5125274');
```

INSERT INTO Movie (Mv_no, Cust_id, Title, Star, Price) VALUES

```
(1, 'A02', 'Bloody', 'JC', 181),  
(2, 'A04', 'The Firm', 'TC', 200),  
(3, 'A01', 'Pretty Woman', 'RG', 151),  
(4, 'A06', 'Home Alone', 'MC', 150),  
(5, 'A05', 'The Fugitive', 'MF', 200),  
(6, 'A03', 'Coma', 'MD', 100),  
(7, 'A02', 'Dracula', 'GO', 150),  
(8, 'A06', 'Quick Change', 'BM', 100),  
(9, 'A03', 'Gone with the Wind', 'CB', 200),  
(10, 'A05', 'Carry on Doctor', 'LP', 100);
```

-- 2. Entity Integrity Check (will fail)

```
-- INSERT INTO Customer VALUES (NULL, 'Test', 'User', 'XX', '1234567');
```

-- 3. Referential Integrity Check (will fail)

```
-- INSERT INTO Movie VALUES (11, 'A99', 'Invalid Movie', 'XX', 150);
```

-- 4. Domain Integrity Check (will fail)

```
-- INSERT INTO Movie VALUES (12, 'A01', 'BadMovie', 'ZZ', 90);
```

-- 5. Movies price > 100 and < 200

```
SELECT Title FROM Movie  
WHERE Price > 100 AND Price < 200;
```

```

-- 6. Cust_id for movies with stars JC, TC, MC
SELECT DISTINCT Cust_id FROM Movie
WHERE Star IN ('JC', 'TC', 'MC');

-- 7. Customers with 'A' in Area
SELECT * FROM Customer
WHERE Area LIKE '%A%';

-- 8. Movies price <= 180 and Title length = 6
SELECT Title FROM Movie
WHERE Price <= 180 AND LENGTH(Title) = 6;

-- 9. Movies with 10% increment
SELECT Title, Price AS Original_Price,
       ROUND(PPrice * 1.10, 2) AS Incremented_Price
FROM Movie;

-- 10. Customer details in sentence
SELECT CONCAT(Fname, ' ', Lname, ' stays in ', Area,
              ' and his phone number is ', COALESCE(Phone, 'not recorded')) AS Details
FROM Customer;

-- 11. Add NOT NULL to Lname
ALTER TABLE Customer MODIFY Lname VARCHAR(20) NOT NULL;

-- 12. Customers with missing phone
SELECT Fname, Lname FROM Customer
WHERE Phone IS NULL;

-- 13. Update missing phone
UPDATE Customer
SET Phone = '9999999999'
WHERE Phone IS NULL;

-- 14. Unique customer IDs from Movie
SELECT DISTINCT Cust_id FROM Movie;

-- 15. Remove NOT NULL from Star
ALTER TABLE Movie MODIFY Star VARCHAR(10) NULL;

-- 16. Delete row from Customer (will fail if referenced in Movie)
DELETE FROM Customer WHERE Cust_id = 'A01';

-- 17. Delete row from Movie (will succeed)
DELETE FROM Movie WHERE Mv_no = 6;

-- 18. Drop Customer (will fail if Movie exists)
DROP TABLE Customer;

-- 19. Drop Movie (will succeed if no dependencies)
DROP TABLE Movie;

```

```
-- 20. Drop Foreign Key from Movie
-- First, find constraint name:
-- SHOW CREATE TABLE Movie;
-- Example: ALTER TABLE Movie DROP FOREIGN KEY movie_ibfk_1;
```